

## API SPECIFICATION DOCUMENT

### PART-3: REQUEST PROCESSING, VALIDATION & ERROR HANDLING

# School ERP

Spring Boot 3 + Java 21 | PostgreSQL

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## Preamble

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This document is Part-3 of the API Specification Document (ASD) v1.0 for the School ERP System, continuing directly from Part-1: API Foundation and Part-2: Authentication, Authorization & API Security. It defines the request-processing conventions — validation, pagination, filtering, sorting, search, file transfer, error handling, and response consistency — that apply uniformly across every API Group already catalogued in Appendix-K Section 18.

This document elaborates, rather than restates, the brief mentions of these topics already made in ASD Part-1 Section 9 and MDB Part-2 Sections 16-17 — where a standard was already fixed there, this Part narrows it to concrete, per-request conventions without contradiction. It contains no source code, no controller code, no JSON, no API endpoint list, no OpenAPI/Swagger, and no sequence diagrams.

## 21. Request Validation Standards

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Fixes the API-boundary application of the Validation Standards already established in MDB Part-2 Section 16, specifically how validation failures are surfaced to the API caller.

- Every Create/Update request is validated in full before any part of it is persisted — validation does not stop at the first failed field; every failing field is identified and returned together, so a caller can correct all issues in one round-trip rather than discovering them one at a time.
- Field-level validation (Mandatory, Format, Range — MDB Part-2 Section 16) occurs before any Business Rule validation (Section 22 this document) — a request with a basic field-level error never reaches Business Rule evaluation, avoiding a misleading Business Rule error on data that was never structurally valid to begin with.
- Validation rules applied at the API boundary match the Form Fields Mandatory/Validation columns already defined per screen in the relevant UI/UX Screen Specification exactly — the API never enforces a stricter or looser rule than the screen specification already defines for the same logical field.
- Read-only fields (per the UI/UX Screen Specifications' Read Only/Editable column) are rejected if present with a different value in an Update request body, rather than silently ignored — this surfaces a caller's incorrect assumption rather than masking it.

## 22. Business Rule Validation Standards

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Fixes how the 71 Business Rules already catalogued in Appendix-C v1.1 are evaluated and surfaced at the API layer, extending MDB Part-3 Section 24 and DDD Part-3 Section 21.

- Business Rule validation occurs entirely in the Service layer (MDB Part-1 Section 8) after field-level validation (Section 21) passes — no Business Rule is ever evaluated at the Controller layer or via a database constraint (DDD Part-3 Section 21).
- A Business Rule violation returns the Business Rule Violation response category (ASD Part-1 Section 10), carrying a reference to the specific Appendix-C v1.1 rule ID without exposing internal rule-engine implementation detail — a caller can distinguish which rule blocked the request without receiving internal diagnostic information.
- Critical-priority Business Rules (19 of 71, Appendix-C v1.1 Section 2.2) receive no different response contract from other rules — the API response format is uniform regardless of rule priority; priority affects only internal escalation and testing rigor (Appendix-M Section 23), not the API contract.
- A request blocked by a Business Rule requiring elevated approval (e.g., BR-ATT-003, BR-EXM-003, BR-FEE-002) returns a response indicating the approval-required state distinctly from a flat rejection, consistent with the ApprovalRequest pattern already fixed in Appendix-H ENT-SYS-007.

## 23. Pagination Standards

Fixes the concrete pagination convention realizing Appendix-K Section 9 and the mandatory-pagination rule already fixed in MDB Part-3 Section 27.

| Aspect                 | Standard                                                                                                                                                                                                                 |
|------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Applicability          | Mandatory for every Search/List operation capable of returning a high-volume result set (DDD Part-2 Section 11 Transaction Data classification) — not optional for these entities                                        |
| Page Size              | A default and a maximum page size apply to every paginated operation; a request exceeding the maximum is capped, not rejected, exact values remain Client/Product Decision Required                                      |
| Response Metadata      | Current page, total result count, and page size accompany every paginated response, per ASD Part-1 Section 10 — never a bare result array                                                                                |
| Consistency            | The same pagination parameter names and response metadata shape are used identically across all 11 modules' API Groups — no module defines its own pagination convention                                                 |
| Non-Paginated Entities | Master Data entities with inherently small result sets (DDD Part-2 Section 11, e.g., Department, Designation) may omit pagination where already noted as "Not paginated" in the corresponding UI/UX Screen Specification |

## 24. Filtering Standards

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Fixes the concrete filtering convention realizing Appendix-K Section 10.

- Filter parameters use the exact field names already defined per entity in Appendix-G's Attribute Catalogue — a filter parameter name never diverges from its corresponding logical attribute name.
- Filtering is supported preferentially on attributes already flagged Indexed=Y in Appendix-G's Attribute Catalogue, consistent with DDD Part-1 Section 10 — a filter on a non-indexed, high-volume-table column is only added alongside a corresponding index addition, not offered without one.
- The specific filter set per API Group matches the Filters already defined per screen in the relevant UI/UX Screen Specification — the API does not offer a filtering capability the corresponding screen does not expose, and vice versa.
- Multiple simultaneous filters are combined with AND semantics by default; OR semantics, where genuinely required by a specific screen's filter behavior, are called out explicitly rather than assumed.

## 25. Sorting Standards

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Fixes the concrete sorting convention realizing Appendix-K Section 11.

- Every paginated/searchable operation supports sorting by at least the Default Sorting field already specified per screen in the relevant UI/UX Screen Specification, applied automatically when no explicit sort parameter is supplied.
- Sortable fields are limited to those already indexed or explicitly justified per DDD Part-1 Section 10 — sorting by a non-indexed column on a high-volume table is not offered, consistent with the Query Performance Standards already fixed in DDD Part-3 Section 27.
- Multi-field sort (e.g., sort by Status, then by Due Date) is supported using a consistent parameter convention across all API Groups, with sort direction (ascending/descending) specified per field independently.

## 26. Search Standards

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Fixes the concrete search convention realizing Appendix-K Section 12 and Appendix-F NFR-SRCH-001, distinguishing Search from Filtering (Section 24).

- Search operates as free-text/fuzzy matching over the specific fields already identified for indexed search in Appendix-H Section 12 (Student.full\_name, Book.title/author, Employee.full\_name) — Filtering (Section 24) remains the mechanism for exact-match or range criteria over structured attributes.
- Every Search operation meets the  $\leq 2$  second response target (NFR-PERF-004) even at the maximum data-volume scale already fixed in Appendix-F's Scalability NFR series, consistent with NFR-SRCH-001.
- Search results are role-scoped identically to the equivalent List/Read operation for the same entity (ASD Part-2 Section 12) — Search never returns a result a caller could not otherwise Read directly.
- A global/cross-module search (e.g., the Attendance Search, Result Search, and equivalent Search screens already defined across the UI/UX Screen Specifications) aggregates results from multiple API Groups behind a single operation, but each contributing result still passes through that entity's own authorization check (Section 22, ASD Part-2 Section 12) individually.

## 27. File Upload & Download Standards

Fixes the concrete file-transfer convention realizing Appendix-K Section 16 and MDB Part-2 Section 18.

- Upload operations use the standard multipart content type (ASD Part-1 Section 8) exclusively; every upload is validated for type and size and scanned for malicious content (NFR-FUS-001) before the operation is considered successful — a successful upload response is never returned before the scan completes.
- Download operations for generated documents (report cards, receipts, payslips, ID cards — Appendix-I Section 12.2) return the appropriate document content type (Section 8, ASD Part-1) and are always mediated through the same authorization check as the owning entity — no generated document is retrievable via a raw, unauthenticated path.
- Large file downloads (e.g., a batch-generated report card set) support the same background-job pattern already fixed for large report exports (Appendix-F NFR-RPTP-001), returning a completion notification rather than blocking the request indefinitely.
- A failed upload (rejected type, oversized, failed scan) returns the Validation Error response category (ASD Part-1 Section 10) with a reason specific enough for the caller to correct the issue, without revealing internal scanning-engine detail.

## 28. Error Handling Standards

Fixes the API-contract elaboration of the Exception Handling Standards already established in MDB Part-2 Section 17, focused on the caller-visible error structure.

| Error Category          | Caller-Visible Contract                                                                                |
|-------------------------|--------------------------------------------------------------------------------------------------------|
| Validation Error        | Field-level detail sufficient to correct every failing field in one pass (Section 21)                  |
| Business Rule Violation | Rule reference without internal detail (Section 22)                                                    |
| Authorization Failure   | Generic denial, no resource-existence disclosure (ASD Part-2 Section 12)                               |
| Concurrency Conflict    | Indicates the caller's data is stale and must be re-fetched (DDD Part-3 Section 21 optimistic-locking) |
| Rate Limit Exceeded     | Indicates the caller should retry after a delay (ASD Part-2 Section 17)                                |
| System Error            | Fully generic; zero internal detail regardless of caller role, per ASD Part-2 Section 16               |

## 29. API Response Consistency Standards

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Fixes the meta-standard ensuring every API Group's responses remain structurally interchangeable from a client's perspective, closing the loop on ASD Part-1 Section 10.

- Every response — success or error, across all 11 modules' API Groups — uses the identical top-level structure (status indicator, result data, metadata, correlation reference per ASD Part-1 Section 7) with no module-specific variation in shape.
- A field present in one API Group's response for a given logical attribute (e.g., a monetary amount, a date) uses the identical data representation convention as the same logical attribute in every other API Group, consistent with the Data Type Standards already fixed in DDD Part-1 Section 8.
- New API Groups or operations introduced after this document's approval are reviewed against this consistency standard before release (Section 30) — a structurally inconsistent new endpoint is treated as a defect, not a stylistic preference.



## 30. API Review Checklist

Verified for every new or changed API operation, complementing the API Security Review Checklist already fixed in ASD Part-2 Section 20 with request-processing and response-consistency items.

| #  | Checklist Item                                                                                                                         |
|----|----------------------------------------------------------------------------------------------------------------------------------------|
| 1  | Field-level validation (Section 21) matches the corresponding UI/UX Screen Specification's Form Fields exactly                         |
| 2  | Business Rule enforcement (Section 22) occurs only in the Service layer, with the correct rule ID referenced in any violation response |
| 3  | Pagination (Section 23) is present and mandatory for any high-volume-capable list operation                                            |
| 4  | Filter and sort fields (Sections 24-25) are limited to indexed or explicitly-justified columns                                         |
| 5  | Search (Section 26) is distinguished from Filtering and meets the NFR-SRCH-001 performance target                                      |
| 6  | File operations (Section 27) enforce type/size/scan validation before success, and never expose a raw storage path                     |
| 7  | Every error category (Section 28) returns its defined caller-visible contract with no internal detail leakage                          |
| 8  | Response structure (Section 29) matches every other API Group's shape with no ad hoc variation                                         |
| 9  | Every field returned traces to an approved Appendix-G attribute or UI/UX Screen Specification Table Column                             |
| 10 | Documentation (migration record, MDB Part-3 Section 28, or equivalent) references the FR/BR/Screen ID the operation implements         |